SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 2-CONCEPT MAPPING

Course Title:

Design and Analysis of Algorithms

Assessment Tool 2 - Concept Mapping

100

Course Code:

CSA06

CO Assessed:

CO1- Precision (BL1, BL2, BL3)

Assessment:

Weightage:

20% of CIA

Total Marks:

CO1 Statement:

Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)

Student Name:

P.Sanjeev

Section / **Batch**:

AIDS/2023

Instructions to **Students**

- Answer the following question. Total Marks: 100.

Reg. No.:

Date:

• Draw a clear and neat concept map for each question.

• Identify all key concepts **and** arrange them in a logical hierarchy.

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Show meaningful linkages between concepts using arrows and connecting words. Compare related concepts wherever required and justify the relationships clearly.

- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type

Concept Mapping

Focus Area

Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method

SECTION A-CONCEPT MAPPING

Questions

Q1

Code of Conduct

1 Psanjeev (192324231 *(Name/Reg No)* certify that *this submission is my original* work *and* that I *have adhered to the guidelines* specified for *this assessment. I understand that any violation of academic* integrity rules will result *in disciplinary action.*

Signature of the Student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3- Proficient
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature

Clarity

10%

Concept map is clear, neat, visually structured, and easy to interpret

Generally clear with minor readability issues

Level 2- **Developing**

Identifies limited concepts; some are irrelevant or missing

Relationships are weak, unclear, *or* partially incorrect

Some organization but lacks clear hierarchy

Limited integration; mostly isolated concepts

Clarity is reduced; difficult to interpret in parts

Level 1- **Beginning**

Fails to identify essential concepts

No meaningful relationships established

No logical organization

or structure

No integration of literature evidence

Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4- Exemplary	Level 3- Proficient	Level 2- Developing	Level 1- Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	**Course Coordinator**	**Head of Department**
Name :	Name:	Name :
Signature:	Signature:	Signature:
Date :	Date:	Date:

Psanjeev

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& Trace how a reccurrence is solved step by step using substitution.

Recurrance Relation

*Extension Nodes:-

Expand the
Recurrence

Identify the
Pattern

Guess the closed form)

Base cone

verification

↓

Mathematical
Induction

confim the closed-form
solution

↓

Determine Time
complexity

- Guess –> used to predict the closed form solution after
  observation the
  expansion.

"Verification →→ confirm that the guered solution satisfies
  the recurrence.

Justification

Mathematical induction proves that the guessed solution is valid for every input size, ensuring the correctness of the reciovence solution rather than relying only on observation,

Interpretatio
n!

The concept map provides a structured method for solving recarence relations. it begins with the

recurrence,

dentifies patterns through. expansion, forms a hypothesis, verifies it wing induction and finally derives the algorithm time complexity.